Project Plan: Million Song Dataset (MSD)

GROUP MEMBERS:

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1. Dataset Selection

Dataset: Million Song Dataset (MSD)

The Million Song Dataset (MSD) is a research-grade collection of audio features and metadata for ~1 million songs. While the full dataset is 280 GB, the 10K-song subset is ideal for manageable development and testing. The dataset includes various audio features such as tempo, loudness, and timbre, as well as metadata like artist and year. The dataset is available for download at: http://millionsongdataset.com/

2. Scenario Description

The project aims to build a music recommendation system using the Million Song Dataset (MSD). The goal is to use audio features (such as tempo, loudness, timbre) to cluster songs into similar categories, which can be used for recommending songs to users based on their listening behavior.

3. Problem Definition

The problem is to create a system that can automatically classify and recommend songs based on their audio features. We aim to leverage clustering algorithms to group similar songs, and later use this information for personalized song recommendations. This unsupervised learning approach does not require labeled data and will be based on inherent patterns in the audio features.

4. Business Goals

The following are the primary business goals for the project:

1. Improve music discovery by clustering songs into similar categories.

2. Enhance recommendation algorithms by incorporating audio features.

3. Increase user engagement through personalized recommendations.

4. Reduce user churn by offering better music suggestions.

5. Enhance model training efficiency by utilizing the full Million Song Dataset (MSD) for comprehensive analysis and better generalization.

5. Data Dictionary

A. Dependent Attribute (N/A for Unsupervised Learning):

N/A (No target variable required for clustering algorithms)

B. Independent Features:

Metadata & Identification

Field	Type	Description
artist_name	String	Name of the artist (Rick Astley).
title	String	Title of the song.
year	Integer	Release year (1987).
release	String	Album name from which the track was taken.
song_id	String	Unique Echo Nest ID for the song.
track_id	String	Unique Echo Nest ID for the specific audio analysis.
audio_md5	String	Hash code of the audio used for the analysis.

External IDs

Field	Type	Description
artist_mbid	String	MusicBrainz ID for the artist.
artist_playmeid	Integer	Artist ID on playme.com.

release_7digitalid	Integer	Album ID on 7digital.com.
track_7digitalid	Integer	Track ID on 7digital.com.

Artist Statistics & Tags

Field	Type	Description
artist_mbtags	Array(4)	Tags assigned to the artist on MusicBrainz.
artist_mbtags_count	Array(4)	Raw count for each MusicBrainz tag.
artist_terms	Array(12)	Descriptive terms/tags from The Echo Nest.
artist_terms_freq	Array(12)	Frequency of terms (0 to 1).
artist_terms_weight	Array(12)	Importance weight of terms (0 to 1).
similar_artists	Array(100)	List of 100 similar artist IDs.
song_hotttnesss	Float	Popularity score at time of download (0.864).

Audio Characteristics

Field	Type	Description
duration	Float	Total track length in seconds (211.7s).
tempo	Float	Estimated beats per minute (113.359 BPM).
loudness	Float	General loudness in decibels (-7.75 dB).
key	Integer	Estimated musical key (1).
key_confidence	Float	Confidence level of the key estimation (0.324).
mode	Integer	Estimated musical mode (1).
mode_confidence	Float	Confidence level of the mode estimation (0.434).
time_signature	Integer	Number of beats per bar (4).

danceability	Float	Measure of danceability (0.0 = not analyzed).
energy	Float	Measure of energy (0.0 = not analyzed).

Field	Type	Description
bars_start	Array(99)	Start times for the 99 musical bars.
beats_start	Array(397)	Start times for the 397 beats.
sections_start	Array(10)	Start times for the 10 identified song sections.
segments_start	Array(935)	Start times for the 935 distinct musical segments.
segments_pitches	Matrix(935, 12)	Chroma features (tonal content) for each segment.
segments_timbre	Matrix(935, 12)	MFCC-like features (texture/quality) for each segment.
tatums_start	Array(794)	Start times for the 794 "tatums" (smallest rhythmic units).
end_of_fade_in	Float	Timestamp where the initial fade-in ends (0.139s).
start_of_fade_out	Float	Timestamp where the final fade-out begins (198.536s).

C. Prediction Model Type:

Unsupervised Learning (Clustering)